

Additional practice problems

STA304 — Summer 2026

Lecture 3

Q1. Suppose an SRS of size n is drawn from a population of size N .

A student claims that

$$\tilde{y} = \frac{1}{N} \sum_{i \in S} y_i$$

is an unbiased estimator of the population mean.

Determine whether the student's claim is correct. Justify your answer mathematically.

Q2. Suppose an SRS of size n is selected without replacement from a population of size N .

- a) How many possible samples of size n are there?
- b) What is the probability of selecting a particular sample S ?
- c) Let

$$\hat{t}_S = N\bar{y}_S.$$

Prove that $\hat{t}_S$ is an unbiased estimator of the population total

$$t = \sum_{i=1}^N y_i.$$

Q3. Suppose an SRS of size n is selected from a population of size N .

Show that the sampling weight for each sampled unit is equal to N/n .

Interpret this weight in words.

Q4. A researcher takes an SRS of size n from a population of size N . Let $y_i = 1$ if unit i has a characteristic of interest and 0 otherwise.

- a) Write the population proportion p in terms of the y_i 's.
- b) Write an estimator $\hat{p}_S$ for p based on the sample.
- c) Prove that $\hat{p}_S$ is unbiased for p .

Q5. Under SRS, the variance of the sample mean is

$$V(\bar{y}_S) = \frac{\sigma_U^2}{n} \left(1 - \frac{n}{N}\right).$$

- a) What happens to this variance when $n = N$?
- b) Explain why this result makes intuitive sense.

- c) What happens to the variance when $N \rightarrow \infty$ while n remains fixed? Interpret your result.
- Q6.** An SRS of size n is drawn without replacement from a population of size N . Let I_i and I_j be inclusion indicators for two distinct units $i \neq j$.
- a) What is $E[I_i]$?
- b) What is $P(I_i = 1, I_j = 1)$?
- c) Explain why I_i and I_j are not independent under sampling without replacement.
- Q7.** A population contains $N = 5000$ households.
- An SRS of size $n = 200$ yields a sample mean monthly electricity bill of \$145 and a sample variance of 625.
- a) Estimate the population mean bill.
- b) Estimate the variance of the sample mean.
- Q8.** A population contains $N = 1200$ individuals.
- An SRS of size $n = 100$ finds that 28 individuals support a proposed policy.
- a) Estimate the population proportion.
- b) Estimate the variance of the estimator.
- c) Interpret the finite population correction.
- Q9.** Two researchers are estimating the same population mean.
- Researcher A samples $n = 100$ units from a population of size $N = 10\,000$.
- Researcher B samples $n = 100$ units from a population of size $N = 150$.
- a) Compute the finite population correction for each study.
- b) Which estimator has the smaller sampling variance?
- c) Explain why this result makes intuitive sense.
- Q10.** A university contains $N = 40\,000$ students.
- An SRS of size $n = 500$ estimates the average weekly spending on food to be

$$\bar{y}_S = 82$$

with sample variance

$$s^2 = 225.$$

Construct a 95% confidence interval for the population mean.

Q11. A company has $N = 2500$ employees.

An SRS of size $n = 150$ estimates the average annual charitable donation to be \$410 with sample variance $s^2 = 3600$.

Construct a 90% confidence interval for the population total charitable donation.

Q12. Researcher A takes an SRS of size 100 from a population of size 500.

Researcher B takes an SRS of size 100 from a population of size 10 000.

Assume both populations have the same variance.

a) Compute the finite population correction for each study.

b) Which estimator has smaller variance?

Q13. A population contains $N = 800$ units.

How large must an SRS be to estimate a population mean with margin of error at most 5 at the 95% confidence level if the population variance is believed to be

$$\sigma_U^2 = 400?$$

Assume asymptotic normality.

Lecture 4

Q1. A population is divided into three strata:

$$N_1 = 500, \quad N_2 = 300, \quad N_3 = 200.$$

A stratified sample of size $n = 100$ is selected using proportional allocation.

a) Compute the sample size allocated to each stratum.

b) What is the inclusion probability for a unit in each stratum?

Q2. A stratified sample is drawn from a population with two strata.

$$N_1 = 400, \quad N_2 = 600$$

The sample means are

$$\bar{y}_{1S} = 25, \quad \bar{y}_{2S} = 40.$$

a) Estimate the population mean.

b) Estimate the population total.

Q3. A stratified sample was selected with

$$N_1 = 1000, \quad N_2 = 2000$$

and

$$n_1 = 100, \quad n_2 = 200.$$

The within-stratum sample variances are

$$s_1^2 = 36, \quad s_2^2 = 49.$$

- a) Estimate the variance of the stratified estimator of the population mean.
- b) Compute the estimated standard error.

Q4. Suppose a population contains

$$N_1 = 3000 \text{ urban households}$$

and

$$N_2 = 1000 \text{ rural households.}$$

You plan to take a stratified sample of size $n = 400$ using proportional allocation.

- a) Determine n_1 and n_2 .
- b) What are the sampling weights in each stratum?
- c) Interpret the sampling weight for a sampled rural household.

Q5. A population is divided into three strata.

$$N_1 = 2000, \quad N_2 = 5000, \quad N_3 = 3000.$$

The within-stratum standard deviations are

$$\sigma_1 = 10, \quad \sigma_2 = 30, \quad \sigma_3 = 20.$$

You can afford a sample of size $n = 400$.

- a) Determine the proportional allocation.
- b) Determine the Neyman (optimal) allocation.

- c) Which stratum gains the largest increase in sample size under optimal allocation? Explain why.

Q6. A stratified sample was selected from two strata.

$$N_1 = 800, \quad N_2 = 1200$$

$$n_1 = 80, \quad n_2 = 120$$

The estimated proportions of support for a policy are

$$\hat{p}_{1S} = 0.30, \quad \hat{p}_{2S} = 0.60.$$

- Estimate the overall population proportion supporting the policy.
- Estimate the total number of people supporting the policy.
- Explain why the overall estimate is not simply the average of 0.30 and 0.60.

Q7. Consider the stratified estimator

$$\bar{y}_{str} = \sum_{h=1}^H \frac{N_h}{N} \bar{y}_{hS}.$$

- What is the expected value of $\bar{y}_{hS}$. You can use results of SRS.
- Use this result to prove that $\bar{y}_{str}$ is unbiased for the population mean.

Q8. A researcher creates strata by randomly assigning units to groups before sampling.

- Will stratification likely improve precision compared to SRS? Explain.
- Describe the type of strata that generally lead to the largest gains in precision.

Q9. A university wishes to estimate average weekly study hours among students.

The population consists of:

- First-year students
- Upper-year students
- Graduate students

- Explain one reason why stratified sampling may be preferable to SRS.
- Propose a stratification variable and justify your choice.
- Explain why the chosen strata may improve precision.

Q10. A public health agency wants to estimate influenza vaccination rates in a province.

The agency is particularly interested in obtaining reliable estimates for seniors aged 65+.

- a) Explain why an SRS may be inadequate.
- b) Describe a stratified sampling design.
- c) Would proportional allocation or optimal allocation be more appropriate if the primary goal is estimating vaccination rates among seniors? Justify your answer.

Q11. A stratified sample is taken from three strata.

$$N_1 = 1000, \quad N_2 = 2000, \quad N_3 = 3000$$

The sample proportions supporting a new policy are

$$\hat{p}_{1S} = 0.20, \quad \hat{p}_{2S} = 0.45, \quad \hat{p}_{3S} = 0.60.$$

- a) Estimate the overall population proportion supporting the policy.
- b) Estimate the total number of individuals supporting the policy.
- c) A student claims the estimate should simply be

$$\frac{0.20 + 0.45 + 0.60}{3}.$$

Explain why this is incorrect.

Lecture 5 + 6

Q1. A researcher takes an SRS of $n = 50$ households from a population of $N = 1000$ households. Let y_i be monthly electricity usage and x_i be household square footage. The population mean square footage is known to be $\bar{x}_{\mathcal{U}} = 1500$.

In the sample, $\bar{y}_S = 820$ and $\bar{x}_S = 1600$.

- a) Compute the estimated ratio $\hat{B}$.
- b) Compute the ratio estimate $\hat{y}_r$ of mean electricity usage.

Q2. A simple random sample of $n = 20$ farms is selected from a population of $N = 500$ farms. Let y_i be crop yield and x_i be farm acreage. The population mean acreage is known.

- a) Explain why acreage might be a useful auxiliary variable for estimating crop yield.
- b) State one condition under which ratio estimation would be more appropriate than the usual SRS sample mean.

- Q3.** A sample of $n = 40$ stores is selected from a population of $N = 400$ stores. The goal is to estimate total annual sales t_y . Let x_i be last year's annual sales and y_i be this year's annual sales.

The known population total of last year's sales is $t_x = 8,000,000$. In the sample,

$$\sum_{i \in S} y_i = 900,000, \quad \sum_{i \in S} x_i = 850,000.$$

- a) Compute $\hat{B}$.
 - b) Compute the ratio estimate $\hat{t}_{yr}$.
 - c) Interpret the estimate in context.
- Q4.** A researcher wants to estimate the average number of weekly transit trips per student. Let y_i be the number of weekly transit trips and x_i be the distance from campus in kilometres. The population mean distance from campus is known to be $\bar{x}_{\mathcal{U}} = 6$ km.

In an SRS of $n = 100$ students, $\bar{y}_S = 8$ and $\bar{x}_S = 5$.

- a) Compute the ratio estimate $\hat{y}_r$.
 - b) Is the ratio estimate larger or smaller than $\bar{y}_S$?
 - c) Explain why this adjustment makes sense if distance from campus and transit use are positively related.
- Q5.** An auditor wants to estimate the ratio of actual inventory value to recorded inventory value. An SRS of $n = 25$ items is selected from a population of $N = 500$ items.

In the sample,

$$\bar{y}_S = 96, \quad \bar{x}_S = 100.$$

- a) Compute $\hat{B}$.
 - b) Interpret $\hat{B}$ in context.
 - c) Explain why estimating a ratio may be more meaningful than estimating the total actual value alone.
- Q6.** A researcher samples $n = 60$ plots from a population of $N = 600$ plots to estimate mean biomass. Let y_i be biomass and x_i be canopy cover. The population mean canopy cover is known to be $\bar{x}_{\mathcal{U}} = 30$.

The sample gives

$$\bar{y}_S = 75, \quad \bar{x}_S = 25.$$

- a) Compute $\hat{B}$.
- b) Compute $\hat{y}_r$.

c) Compare $\hat{y}_r$ to $\bar{y}_S$.

d) Explain what the adjustment implies about the sampled plots relative to the population.

Q7. An SRS of $n = 50$ units is drawn from a population of $N = 1000$. A ratio estimator is used to estimate $\bar{y}_U$.

Suppose

$$\bar{x}_U = 40, \quad \bar{x}_S = 50, \quad \bar{y}_S = 120,$$

and

$$s_e^2 = \frac{1}{n-1} \sum_{i \in S} (y_i - \hat{B}x_i)^2 = 36.$$

a) Compute $\hat{y}_r$.

b) Estimate the sampling variance of $\hat{y}_r$ using

$$\hat{V}(\hat{y}_r) = \left(1 - \frac{n}{N}\right) \left(\frac{\bar{x}_U}{\bar{x}_S}\right)^2 \frac{s_e^2}{n}.$$

c) Construct an approximate 95% confidence interval for $\bar{y}_U$.

Q8. A survey is conducted to estimate total household spending in a city. An SRS of $n = 100$ households is selected from $N = 10,000$ households. Let x_i be household income and y_i be household spending.

The known population total income is

$$t_x = 600,000,000.$$

The sample totals are

$$\sum_{i \in S} y_i = 5,800,000, \quad \sum_{i \in S} x_i = 6,200,000.$$

The residual variance is $s_e^2 = 4000$.

a) Compute $\hat{B}$.

b) Compute the ratio estimate $\hat{t}_{yr}$.

c) Estimate the sampling variance of $\hat{t}_{yr}$ using

$$\hat{V}(\hat{t}_{yr}) = \left(1 - \frac{n}{N}\right) \left(\frac{t_x}{\bar{x}_S}\right)^2 \frac{s_e^2}{n}.$$

d) Explain why the variance formula depends on s_e^2 rather than s_y^2 .

- Q9.** A researcher is deciding whether to use the usual SRS estimator $\bar{y}_S$ or the ratio estimator $\hat{y}_r$.

Suppose the population correlation between x and y is $R = 0.75$, the coefficient of variation of y is $CV(y) = 0.40$, and the coefficient of variation of x is $CV(x) = 0.60$.

- a) Compute

$$R \frac{CV(y)}{CV(x)}.$$

- b) Based on the condition

$$R \frac{CV(y)}{CV(x)} > \frac{1}{2},$$

determine whether the ratio estimator is expected to have smaller MSE than the usual SRS estimator.

- c) Explain the intuition behind this condition in words.

- Q10.** A researcher wants to estimate the total number of fish longer than 12 cm in a haul. The total weight of the haul is known, but the total number of fish in the haul is unknown.

Let

$$y_i = \begin{cases} 1, & \text{if fish } i \text{ is longer than 12 cm,} \\ 0, & \text{otherwise,} \end{cases}$$

and let x_i be the weight of fish i .

- a) Explain why the usual estimator $N\bar{y}_S$ cannot be used directly.
- b) Write an estimator of the unknown population size N using the known total weight t_x and the sample mean weight $\bar{x}_S$.
- c) Use this to write an estimator of the total number of fish longer than 12 cm.
- d) Explain why this is an example of ratio estimation.
- Q11.** A simple random sample of n units is selected from a population of size N . Let x_i be an auxiliary variable with known population mean $\bar{x}_U$.

The ratio estimator for the population mean is

$$\hat{y}_r = \hat{B}\bar{x}_U, \quad \hat{B} = \frac{\bar{y}_S}{\bar{x}_S}.$$

- a) Explain why $\hat{y}_r$ is generally biased.
- b) State two conditions under which the bias is expected to be small.
- c) Explain why the proportionality assumption $y_i \approx Bx_i$ is important.

- d) Give an example where regression estimation would likely be preferred over ratio estimation.

Q12. A researcher samples $n = 80$ businesses from a population of $N = 2000$ businesses to estimate total current revenue. Last year's total revenue for all businesses is known.

A colleague suggests using ratio estimation because last year's revenue and current revenue are highly correlated.

- a) Explain why a high correlation between x and y is helpful for ratio estimation.
- b) Explain why high correlation alone is not always sufficient for ratio estimation to be appropriate.
- c) Describe what feature of the relationship between x and y would make ratio estimation especially appropriate.
- d) Describe what feature of the relationship would make regression estimation more appropriate.

Q13. A population follows the exact relationship

$$y_i = 10 + 2x_i.$$

An SRS is taken, and the population mean $\bar{x}_{\mathcal{U}}$ is known.

- a) Does the relationship between x and y pass through the origin? Explain.
- b) Would ratio estimation be exactly appropriate here? Explain.
- c) Would regression estimation be more appropriate here? Explain.
- d) Explain how the nonzero intercept can create bias in a ratio estimator, even when the relationship between x and y is perfectly linear.

Q14. A researcher wants to estimate the total value of equipment in a company. The book value x_i is known for every item, but the actual value y_i must be measured by an auditor. An SRS of $n = 100$ items is selected from $N = 5000$ items.

The researcher uses

$$\hat{t}_{yr} = \hat{B}t_x, \quad \hat{B} = \frac{\bar{y}_S}{\bar{x}_S}.$$

- a) Interpret $\hat{B}$ in this setting.
- b) Explain why t_x is useful auxiliary information.
- c) Explain why the ratio estimator calibrates the sample estimate using the known total book value.

- d) Suppose $\bar{x}_S$ is larger than $\bar{x}_U$ and actual value is positively associated with book value. Should the ratio estimate be larger or smaller than the usual SRS estimate? Explain.
- e) Explain why ratio estimation can reduce variance but may introduce bias.

Lecture 8.2

Q1. A transit agency wishes to estimate the total number of passengers using all bus routes on a particular day.

The agency has four routes.

Route	Daily Scheduled Trips (M_i)	Passengers (t_i)
A	10	400
B	20	780
C	30	1180
D	40	1610

One route is selected with probability proportional to the number of scheduled trips.

- Compute the selection probability of Route B.
- Compute the sampling weight of Route B.
- Suppose Route B is selected. Estimate the total number of passengers.
- Explain why the number of scheduled trips might be correlated with passenger counts.
- Explain why PPS sampling may lead to a smaller variance than selecting routes with equal probabilities.

Q2. A city contains four public libraries with the following numbers of visitors per month:

(200, 500, 700, 1600).

A researcher wishes to estimate the total number of books currently on loan and plans to sample one library.

- Explain why selecting libraries with equal probability may be inefficient.
- Suggest a reasonable measure of size for unequal-probability sampling and justify your choice.

Q3. A survey selects one restaurant using probability proportional to seating capacity.

The selected restaurant has

$$\psi_i = 0.08, \quad t_i = 320$$

where t_i is the total number of customers served on a particular day.

- a) Compute the estimate of the population total.
- b) Explain why the estimate is much larger than the observed value.
- c) If a restaurant with $\psi_i = 0.25$ and the same value of t_i had been selected, would the estimate be larger or smaller? Explain without calculation.

Q4. A city contains several apartment buildings. A sample of $n = 3$ buildings is selected with replacement using probabilities proportional to the number of apartments.

The sampled buildings are shown below.

Building	ψ_i	t_i
4	0.10	85
7	0.20	160
2	0.05	45

Here t_i denotes the total number of residents in building i .

- a) Estimate the total number of residents in the population.
- b) Estimate the standard error of your estimator.

Q5. A university samples $n = 4$ classes with replacement using unequal probabilities.

For each sampled class, M_i is the class size and t_i is the total number of hours studied by students in that class during the previous week.

Class	ψ_i	M_i	t_i
12	0.10	30	120
5	0.08	25	100
5	0.08	25	100
18	0.12	40	180

- a) Estimate the population total number of study hours.
- b) Estimate the population size M_0 .
- c) Estimate the average number of study hours per student.

Q6. A fisheries agency samples fishing boats with replacement using unequal probabilities.

The following information is collected.

Boat	ψ_i	M_i	t_i
3	0.06	40	720
8	0.12	65	1170
11	0.08	50	850

Here M_i is the number of traps hauled and t_i is the total weight (kg) of crab caught.

- a) Estimate the total crab catch in the population.
- b) Estimate the total number of traps in the population.
- c) Estimate the average kilograms of crab caught per trap.

Q7. A researcher samples hospitals with replacement using probabilities proportional to the number of beds.

The sampled hospitals produce the following values:

$$\frac{t_i}{\psi_i} = (950, 1100, 1050, 1000, 900).$$

- a) Estimate the population total.
- b) Estimate the standard error.
- c) Based on the estimated standard error, comment on the precision of the estimate.

Q8. Suppose one PSU is selected from a population of N PSUs using unequal probabilities $\psi_1, \dots, \psi_N$.

Let

$$\hat{t}_\psi = \frac{t_i}{\psi_i}$$

if PSU i is selected.

Show that $\hat{t}_\psi$ is an unbiased estimator of the population total

$$t = \sum_{i=1}^N t_i.$$

Be sure to justify each step in your derivation.

Q9. Suppose a single PSU is selected with probabilities $\psi_1, \dots, \psi_N$.

Define

$$I_i = \begin{cases} 1, & \text{if PSU } i \text{ is selected} \\ 0, & \text{otherwise.} \end{cases}$$

The estimator of the population total is

$$\hat{t}_\psi = \sum_{i=1}^N I_i \frac{t_i}{\psi_i}.$$

- a) Find $E[I_i]$.
- b) Use your answer to show that $\hat{t}_\psi$ is unbiased for t .

Q10. A sample of n PSUs is selected with replacement using selection probabilities $\psi_1, \dots, \psi_N$.
Let

$$u_j = \frac{t_j}{\psi_j}$$

denote the contribution from the PSU selected on draw j .

The Hansen–Hurwitz estimator is

$$\hat{t}_\psi = \frac{1}{n} \sum_{j=1}^n u_j.$$

- a) Find $E[u_j]$.
- b) Use your answer to show that $E[\hat{t}_\psi] = t$.

Q11. Suppose a sample of n PSUs is selected with replacement using unequal probabilities.
Let

$$Q_i$$

denote the number of times PSU i appears in the sample.

- a) What is the distribution of Q_i ?
- b) Find $E[Q_i]$.
- c) Explain the practical interpretation of $E[Q_i]$ in the context of unequal-probability sampling.

Q12. In unequal-probability sampling with replacement, suppose PSU i has selection probability $\psi_i = 0.20$ and the sample size is $n = 50$.

- a) Find the expected number of times PSU i appears in the sample.
- b) Explain why a PSU may appear more or fewer times than this expected value in a particular sample.

Lecture 9

- Q1.** Consider the research question: Does participation in a weekly tutoring program improve student grades?
- a) Identify a possible treatment group and control group.
 - b) Identify the experimental unit.
 - c) Identify a suitable response variable.
 - d) Explain why this question is about a causal effect.
- Q2.** A researcher wishes to determine whether a new sleep medication improves sleep quality.
- a) Explain why it is impossible to observe both potential outcomes for the same individual.
 - b) Describe how a randomized experiment can be used to estimate the causal effect.
 - c) Explain why comparing treatment and control groups can provide evidence of causation.
 - d) Explain why an observational study would generally provide weaker evidence.
- Q3.** A study compares people who choose to exercise outdoors with people who choose not to exercise outdoors. The outdoor group reports higher sleep quality.
- a) Explain why this study is observational rather than experimental.
 - b) Define a confounder.
 - c) Give one plausible confounder in this study.
 - d) Explain how the confounder could distort the observed relationship.
- Q4.** In an experiment on daylight exposure and sleep quality, participants are allowed to choose whether they will spend one hour outdoors each day.
- a) Explain why self-selection may create differences between treatment groups.
 - b) Explain why physically active individuals may be more likely to choose the daylight exposure group.
 - c) Explain why physical activity could be considered a confounder.
 - d) Explain why the resulting comparison cannot be interpreted as a causal effect of daylight exposure alone.
- Q5.** Suppose participants are randomly assigned to either a daylight exposure group or a no-exposure group.

- a) Explain how randomization helps balance confounders.
 - b) Explain why randomization helps isolate the effect of the treatment.
 - c) Does randomization guarantee perfectly identical groups? Explain.
 - d) Explain why randomization strengthens causal conclusions.
- Q6.** A survey asks respondents about their weekly exercise habits. Some respondents skip the exercise question.
- a) Define nonresponse.
 - b) Explain how nonresponse can introduce bias.
 - c) Explain why a high response rate does not automatically guarantee low bias.
 - d) Describe one strategy that could be used to reduce nonresponse.
- Q7.** In a student wellness survey, the probability of responding depends on age. Age is observed for all sampled students.
- a) Explain why this situation may be classified as MAR.
 - b) State the key assumption required for MAR.
 - c) Explain how age information might be used to adjust for nonresponse.
 - d) Explain why bias may remain if the MAR assumption is violated.
- Q8.** A survey experiences missing responses on an income question. Individuals with very high incomes are less likely to respond, even after accounting for all observed variables.
- a) Classify the missingness mechanism.
 - b) Explain why it falls into this category.
 - c) Explain why adjustment based only on observed variables may be insufficient.
 - d) Describe the potential impact on estimates of average income.
- Q9.** Compare the following missing data mechanisms.
- a) Define MCAR.
 - b) Define MAR.
 - c) Define NMAR.
 - d) Which mechanism generally poses the greatest challenge for survey inference? Explain.

- Q10.** A survey is conducted using an SRS of adults. Administrative records contain age group information for every sampled individual, including nonrespondents.
- a) Explain the rationale behind the two-stage approach to nonresponse adjustment.
 - b) Describe the role of response probabilities in the adjustment process.
 - c) Explain how weighting can be used to compensate for nonresponse.
 - d) Explain why the resulting estimator may still be biased.
- Q11.** A survey analyst estimates response rates separately for three age groups and then inflates respondent weights by the inverse of the estimated response rate.
- a) Explain the intuition behind inverse-response weighting.
 - b) Why do respondents in low-response groups receive larger weights?
 - c) Under what assumptions can this reduce nonresponse bias?
 - d) What tradeoff may occur when using highly variable weights?
- Q12.** Explain the difference between an observational study and a randomized experiment.
- a) Which design is generally preferred for studying causal effects?
 - b) Why does the preferred design provide stronger evidence?
 - c) Give an example of a research question that could reasonably be studied using an observational study.
 - d) Give an example of a research question that would be well suited to a randomized experiment.
- Q13.** A survey selected $n = 200$ individuals.

After the initial survey:

$$n_R = 150, \quad n_M = 50.$$

A second phase of sampling is conducted among the nonrespondents. A sample of 20 of the 50 nonrespondents is contacted again and all respond.

The respondent mean is

$$\bar{y}_R = 30$$

and the mean among the second-phase sample of nonrespondents is

$$\bar{y}_M = 42.$$

- a) What proportion of the nonrespondents was sampled in the second phase?
- b) Compute the two-phase estimate of the population mean.
- c) Why might the two-phase estimate differ from the respondent-only estimate $\bar{y}_R$?
- d) Briefly explain why the two-phase approach may reduce nonresponse bias.

Q14. A survey selected an SRS of $n = 500$ individuals to estimate average monthly spending on public transportation.

After the initial survey:

$$n_R = 350, \quad n_M = 150.$$

The survey team decides to use a two-phase sampling approach and contacts a random sample of 60 of the nonrespondents by telephone. All 60 provide data.

The respondent mean is

$$\bar{y}_R = 85$$

and the mean among the second-phase sample of nonrespondents is

$$\bar{y}_M = 110.$$

- a) Compute the two-phase estimate of the population mean.
- b) Compare your answer to the respondent-only estimate. Which estimate is larger?
- c) What does the difference between $\bar{y}_R$ and $\bar{y}_M$ suggest about the potential for nonresponse bias?
- d) Explain why a second phase of sampling nonrespondents may improve the estimate.

Q15. A survey selected $n = 1000$ individuals.

There were $n_R = 800$ respondents and $n_M = 200$ nonrespondents.

A second phase sample of 40 nonrespondents is selected.

- a) What percentage of the nonrespondents was selected in the second phase?
- b) Each sampled nonrespondent in the second phase represents how many originally sampled nonrespondents?
- c) Why might researchers choose to conduct a second phase of sampling rather than ignore the nonrespondents?

- d) Name one way the second phase could differ from the first phase of data collection.

Lecture 10

Q1. A survey uses weighting class adjustments to account for nonresponse.

- a) What is a weighting class?
- b) Explain how the weight of a nonrespondent is handled within a weighting class.

Q2. Suppose an SRS of $n = 100$ is selected from a population of $N = 1000$.

Within a weighting class, 25 sampled units were selected and 20 responded.

- a) What is the original sampling weight?
- b) Find the nonresponse-adjusted weight for respondents in this weighting class.

Q3. A researcher is creating weighting classes based on age group.

- a) Why is age a potentially useful variable for creating weighting classes?
- b) State one condition under which weighting class adjustments will produce approximately unbiased estimators.

Q4. A survey has known population totals for three age groups.

The researcher decides to use post-stratification.

- a) What information is required to perform post-stratification?
- b) Explain the goal of modifying weights using post-stratification.

Q5. A survey measures sex (male/female) and race (five categories).

Population totals are known for sex and race separately, but not for every sex-by-race combination.

- a) What adjustment method would be appropriate in this situation?
- b) Explain why ordinary post-stratification is difficult to apply.

Q6. Consider the raking procedure.

- a) After adjusting the rows to match known row totals, what problem can occur?
- b) Briefly describe how raking addresses this problem.

Q7. A survey has missing values for annual income.

A researcher replaces each missing value with the sample mean income.

- a) What type of imputation is being used?
- b) State one advantage and one disadvantage of this approach.

Q8. Suppose a weighting class contains respondents and nonrespondents.

a) Why do we check that

$$\sum_{i \in \mathcal{R}} \tilde{\omega}_i = \sum_{i \in \mathcal{S}} \omega_i?$$

b) What does this equality tell us about the redistribution of weights?

Q9. An SRS of $n = 200$ is selected from a population of $N = 2000$.

Within the age 18–34 weighting class, 80 people were sampled and 64 responded.

a) Find the original sampling weight.

b) Estimate the response propensity for this weighting class.

c) Find the nonresponse-adjusted weight for respondents in this class.

Q10. A weighting class contains sampled units with a total sampling weight of 1200.

The respondents in the class have a total sampling weight of 900.

a) Compute the estimated response propensity $\hat{\phi}_c$.

b) Compute the weighting-class adjustment factor.

c) Find the adjusted weight for a respondent whose original weight was 8.

Q11. The current weighted total for females is 1620.

The known population total is $N_{\text{Female}} = 1510$.

A female respondent currently has weight 20.

a) Compute the raking adjustment factor for females.

b) Compute the respondent's updated weight after the row adjustment.

Q12. An SRS of $n = 100$ is selected from a population of $N = 1000$.

There are two weighting classes.

Class	Respondents	Adjusted Weight
1	30	12
2	40	16

The mean outcome among respondents is 50 in Class 1 and 80 in Class 2.

a) Estimate the population total using weighting-class adjustment.